Chervith Nannuru

nannuruchervith@gmail.com — LinkedIn — GitHub — Portfolio

Education

Mahindra University Aug 2023 – Present
B.Tech in Computer Science CGPA: 8.2/10
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Neural Networks, Database Management Systems, Operating Systems, Object-Oriented Programming.

Skills

Languages: Python, C
Frameworks: PyTorch, TensorFlow, Scikit-learn, FastAPI
Libraries: NumPy, Pandas, OpenCV, Matplotlib
Core Concepts: Deep Learning, CNNs, Transfer Learning, Computer Vision, Transformers
Tools: Git, GitHub, Jupyter Notebook, Google Colab, Vercel
Web: HTML, CSS, JavaScript

Projects

Predicting Human Annotator Disagreement Jan 2026 – Feb 2026
GitHub

- Designed a two-stage pretraining and fine-tuning pipeline to predict soft probability distributions over human annotator disagreement on the CIFAR-10H dataset.
- Implemented and evaluated a custom CNN and ResNet-20 using KL Divergence, Soft Cross-Entropy, and a custom KL+Entropy objective for distribution-based learning.
- Achieved the best performance with ResNet-20, obtaining **0.9133** Cosine Similarity and **0.3273** KL Divergence on the evaluation set.
- Modeled annotator uncertainty by learning soft probability distributions instead of conventional one-hot labels.

LLM from Scratch Jun 2026 – Jul 2026
GitHub

- Built a decoder-only GPT model from scratch in PyTorch by implementing token embeddings, positional embeddings, and stacked transformer blocks.
- Implemented transformer blocks with multi-head self-attention, feed-forward networks, residual connections, layer normalization, and causal masking.
- Loaded pretrained GPT-2 (124M) weights into the implemented GPT model and performed autoregressive text generation.

Potato Leaf Disease Classification May 2026 – Jun 2026
GitHub — Live Demo

- Built an end-to-end potato leaf disease classification system using TensorFlow and ResNet50 transfer learning.
- Applied preprocessing, augmentation, and fine-tuning on the PlantVillage dataset to classify three disease categories.
- Achieved **98%+** validation accuracy and validated predictions on unseen leaf images.
- Developed a FastAPI inference backend, deployed the API on Render and the frontend on Vercel, enabling real-time potato leaf disease prediction through a web application.

Awards & Achievements

- Top 10 finish among 80 teams in the Autonomous Drone Development Competition (ADDC).
- Top 5 finish in the Mahindra University 24-hour Hackathon.

Leadership

Software Vice Head, AERO Club, Mahindra University

- Led software development activities for autonomous drone competitions.
- Mentored junior members on DroneKit and ArduPilot workflows.
- Coordinated technical workshops and software sessions for club members.